

Randomized experiments

INFO/STSCI/ILRST 3900: Causal Inference

1 Sep 2026

Learning goals for today

At the end of class, you will be able to:

1. Explain why exchangeability holds in randomized experiments
2. Understand why exchangeability allows for direct estimation of causal effects

Logistics

- ▶ Bring laptop to discussion section tomorrow

Potential outcome notation review

- ▶ We typically use i to denote a generic unit in our study
- ▶ Y_i^a denotes the outcome we would have observed for unit i if assigned treatment value a
e.g., $a = \text{MedDiet}$ or $a = \text{PruDiet}$
- ▶ Y_i is the observed outcome for unit i
- ▶ A_i is treatment received by unit i

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e.g., $a = \text{MedDiet}$ or $a = \text{PruDiet}$
- ▶ Y_i is the observed outcome for unit i
- ▶ A_i is treatment received by unit i
- ▶ The causal effect for individual i can be defined as

$$Y_i^{\text{treatment}} - Y_i^{\text{NoTreatment}}$$

- ▶ Fundamental problem of causal inference is that we don't observe counterfactual outcomes

Randomized experiments

The New York Times

TheUpshot

What Coronavirus Researchers Can Learn From Economists

Randomized controlled trials remain the gold standard, but natural experiments can help doctors who need answers now.



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By Anupam B. Jena and Christopher M. Worsham

June 30, 2020



Fundamental problem of causal inference

- ▶ Randomized experiments are the gold standard for estimating causal effects
- ▶ Data is still missing in random experiments

	A (vaccine status)	Y (covid status)	$Y^{a=1}$	$Y^{a=0}$	$Y^{a=1} - Y^{a=0}$
Ind 1	0	0	?	0	?
Ind 2	0	1	?	1	?
Ind 3	0	0	?	0	?
Ind 4	1	1	1	?	?
Ind 5	1	0	0	?	?
Ind 6	1	1	1	?	?

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Why do randomized experiments “work”?

What can go wrong?

Suppose we tested all individuals in Ithaca in the Fall of 2021

- ▶ Whether the individual was vaccinated for Covid
 $A_i = 1$ if vaccinated, $A_i = 0$ if not vaccinated
- ▶ Whether the individual tested positive for Covid in 2021
 $Y_i = 1$ if positive test, $Y_i = 0$ if no positive test

What can go wrong?

Front of class

- ▶ Of the **vaccinated** individuals ($A_i = 1$), 50% had a positive Covid test ($Y_i = 1$)

$$E(Y \mid A = 1) = .5$$

- ▶ Of the **not vaccinated** individuals ($A_i = 0$), 70% had a positive Covid test ($Y_i = 1$)

$$E(Y \mid A = 0) = .7$$

- ▶ How might a vaccine skeptic explain the data?

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Back of class

- Of the individuals who are **vaccinated** ($A_i = 1$), 70% had a positive Covid test ($Y_i = 1$)

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- Of the individuals who are **not vaccinated** ($A_i = 0$), 50% had a positive Covid test ($Y_i = 1$)

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- How might a vaccine advocate explain the data?

Randomized experiment

Experiment designed by Pfizer **randomly assign** each individual (43,000 total) into two groups¹:

- ▶ Two doses 21 days apart of either:
BNT162b2 vaccine ($A_i = 1$) or placebo ($A_i = 0$)
- ▶ Whether a positive Covid test was recorded between 7 days and 14 weeks after the injection

¹Polack et. al. NEJM 2020

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- ▶ Whether a positive Covid test was recorded between 7 days and 14 weeks after the injection
- ▶ Of the individuals who were given the vaccine ($A_i = 1$), 0.04% had a positive Covid test ($Y_i = 1$)
- ▶ Of the individuals who were given the placebo ($A_i = 0$), 0.9% had a positive Covid test ($Y_i = 1$)
- ▶ Individuals who received the placebo were ≈ 20 times more likely to get Covid than individuals who recieved the vaccine

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Do the skeptics objections still hold?

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Why experiments “work”

Table 1. Demographic Characteristics of the Participants in the Main Safety Population.*

Characteristic	BNT162b2 (N=18,860)	Placebo (N=18,846)	Total (N=37,706)
Sex — no. (%)			
Male	9,639 (51.1)	9,436 (50.1)	19,075 (50.6)
Female	9,221 (48.9)	9,410 (49.9)	18,631 (49.4)
Race or ethnic group — no. (%)†			
White	15,636 (82.9)	15,630 (82.9)	31,266 (82.9)
Black or African American	1,729 (9.2)	1,763 (9.4)	3,492 (9.3)
Asian	801 (4.2)	807 (4.3)	1,608 (4.3)
Native American or Alaska Native	102 (0.5)	99 (0.5)	201 (0.5)
Native Hawaiian or other Pacific Islander	50 (0.3)	26 (0.1)	76 (0.2)
Multiracial	449 (2.4)	406 (2.2)	855 (2.3)
Not reported	93 (0.5)	115 (0.6)	208 (0.6)
Hispanic or Latinx	5,266 (27.9)	5,277 (28.0)	10,543 (28.0)
Country — no. (%)			
Argentina	2,883 (15.3)	2,881 (15.3)	5,764 (15.3)
Brazil	1,145 (6.1)	1,139 (6.0)	2,284 (6.1)
South Africa	372 (2.0)	372 (2.0)	744 (2.0)
United States	14,460 (76.7)	14,454 (76.7)	28,914 (76.7)
Age group — no. (%)			
16–55 yr	10,889 (57.7)	10,896 (57.8)	21,785 (57.8)
>55 yr	7,971 (42.3)	7,950 (42.2)	15,921 (42.2)
Age at vaccination — yr			
Median	52.0	52.0	52.0
Range	16–89	16–91	16–91
Body-mass index‡			
≥30.0: obese	6,556 (34.8)	6,662 (35.3)	13,218 (35.1)

* Percentages may not total 100 because of rounding.

† Race or ethnic group was reported by the participants.

‡ The body-mass index is the weight in kilograms divided by the square of the height in meters.

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Because treatment is randomly assigned, the distribution of **potential outcomes** for those who are treated and those who are not treated (control group) are similar!

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This is a concept called **exchangeability**

Sometimes also referred to as **exogenous**

$\underbrace{exo}_{\text{outside}} \cdot \underbrace{genous}_{\text{to produce}}$

Exchangeability

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- Knowing $(Y^{a=1}, Y^{a=0})$ does not help me predict A_i
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- Knowing $(Y^{a=1}, Y^{a=0})$ does not help me predict A_i
- Distribution of $Y^{a=1}$ and $Y^{a=0}$ is same for individuals with $A = 1$ and $A = 0$
- Exchangeability does **not** mean that the **observed** outcome Y is independent of the observed treatment!

Why is exchangeability good?

The average treatment effect (ATE)² is the difference in average outcome that would occur if everyone is treated compared to the average outcome that would occur if no-one is treated

$$\text{ATE} = E(Y^{a=1} - Y^{a=0}) = \underbrace{E(Y^{a=1})}_{\text{if everyone is treated}} - \underbrace{E(Y^{a=0})}_{\text{if no-one is treated}}$$

²Often also called the average causal effect (ACE)

Why is exchangeability good?

When exchangeability (and consistency) is true, it implies

$$\begin{aligned} \underbrace{E(Y \mid A = 1)}_{\text{What we could observe}} &= \underbrace{E(Y^{a=1} \mid A = 1)}_{\text{Within treated}} && (\textit{Consistency}) \\ &= \underbrace{E(Y^{a=1} \mid A = 0)}_{\text{Within not treated}} && (\textit{Exchangeability}) \\ &= \underbrace{E(Y^{a=1})}_{\text{everyone}} && (\textit{Exchangeability}) \end{aligned}$$

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This allows us to identify the average treatment effect (ATE)

$$\text{ATE} = \underbrace{E(Y^{a=1})}_{\text{if everyone is treated}} - \underbrace{E(Y^{a=0})}_{\text{if no-one is treated}}$$

Identification vs estimation

- ▶ We used expectations which refer to population quantities (i.e., average over infinite data)
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Philosophical strategy: make sure we can identify the quantity of interest first, then worry about estimation

Observational Setting

- ▶ Given your potential outcomes, pick a diet for yourself:
Delicious or Healthy (A_i)
- ▶ Under consistency, what is Y_i ?

	$Y_i = \text{Disease (0)}$	$Y_i = \text{No Disease (1)}$
$A_i = \text{Delicious (0)}$		
$A_i = \text{Healthy (1)}$		

$$E(Y \mid A = 1) - E(Y \mid A = 0) =$$

Experimental Setting

- ▶ If your birthday month is odd: $A_i = \text{Delicious}$
- ▶ If your birthday month is even: $A_i = \text{Healthy}$
- ▶ Under consistency, what is Y_i ?

	$Y_i = \text{Disease (0)}$	$Y_i = \text{No Disease (1)}$
$A_i = \text{Delicious (0)}$		
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$$E(Y \mid A = 1) - E(Y \mid A = 0) =$$

True ATE

- ▶ 50% of students have:

$$Y_i^{\text{Delicious}} = 0, Y_i^{\text{Healthy}} = 1$$

- ▶ 50% of students have:

$$Y_i^{\text{Delicious}} = 1, Y_i^{\text{Healthy}} = 1$$

True ATE

- ▶ 50% of students have:

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- ▶ 50% of students have:

$$Y_i^{\text{Delicious}} = 1, Y_i^{\text{Healthy}} = 1$$

$$ATE = E(Y^{\text{Healthy}}) - E(Y^{\text{Disease}}) = 1 - .5 = .5$$

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